The Fold-Equation Stack

Round 31: the emergence layer, the section theorem, the split-deck formula, the square-zero dressing law, and the integral gauge at $2m$

The campaign (rounds 1–30) and this round's strike

2026-11-02 (Round 31)

Abstract

This document is the complete mathematical record of Round 31, the *fold-equation stack*: the operator's directive — take the fold equations themselves, force the Hodge structure out of them, and exclude the counterexamples — executed at exactly the strength the mathematics supports. The stack, each layer proven and certified (`fold_equation_stack_checks.py`, 36 checks, exit 0, deterministic double-run byte-identical, exact arithmetic, zero floats): **(1) the emergence layer** — every 2-torsion-free lattice carrying $\gamma^2 = -1$ (the fold equation) is canonically a $\mathbb{Z}[i]$-module; the bigrading, the conjugation theorem, the door group $(\mathbb{Z}/2)^s$, the order formula $|\det(\gamma - 1)| = 2^s$, and the parity door R5 are laws of the fold equation alone, not of the committed model (certificate family A, on randomly conjugated $\mathbb{Z}[i]^s$ decks); **(2) the section theorem** — the lifting cocycle c_ν of Round 28 is a law of *any* semidirect C_4-extension with deck-pulled content: the bar identity uses only the semidirect law (family B); **(3) the split-deck formula** — $H^1(C_4; \mathrm{Hom}(\Gamma, A)) \cong (\mathbb{Z}/2)^{ts}$ for split decks, with the pair-block vanishing and the $\mathbb{Z}/4$ glue control (families C, E); **(4) the square-zero dressing law** — the rotation lattice $L = \ker(\Phi^2 + 1)$ has rank 24 with the transparent block law, and every equivariant realization span is square-zero: the dressing $M = I + X(c)$ is a homomorphic lift of the additive cocycle, so the multiplicative two-cell carries *no quadratic obstruction* (family F, with the honest per-span scope receipt); **(5) THE INTEGRAL GAUGE AT $2m$** — the primitive equation $(\gamma_{\mathrm{act}} - 1)\sigma = 2f_w$ has the integral solution $\sigma(v) = v_{f_1}(W_R - W_I)$; the additive two-cell obeys $\delta s = 2c_\nu$; and the doubled dressed transport is exactly gauge-trivial, $\delta S = M_2$ with $S\colon G_\bullet \to \mathrm{SL}_8(\mathbb{Z})$ integral and normalized — the complete 1152-cell basis sweep closing the universal quantifier, uniform over all 24 realizations, with the scale-1 halves obstruction as the hostile receipt that the kill at $2m$ is load-bearing (family G); **(6) the μ_2 layer** — $H^p(C_4; \mu_2) = \mu_2$ for every p and the gerbe ambient $H^3(G_\bullet; \mu_2) = \mathbb{F}_2^{71}$ recomputed genuinely through the wedge matrices of the mod-2 letter deck (family H). The corollary is the round's theorem: **the residual μ_2-class of the doubled dressed transport — the boundary-(a) object that Round 29 named as the next strike target — is computed: zero.** The stratum theorem assembles the stack into the forcing argument: no counterexample exists in the fold-carried stratum; the counterexample locus is confined to the fold-free stratum, the input axis (ledger 085), where the deck-existence itself remains the single open object. Q2 is not claimed; the closing string is not emitted.

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Disposition (on the face)

- **Q1 (the framework theorem, $\mathbf{FS}_\forall$ on the committed world model)**: *proven and certified*, CI-green (unchanged; Round 30 state). This round extends the certified chain *through the multiplicative layer*: the nonabelian two-cell of the doubled dressed transport is integrally gauge-trivial (Theorem 6.4), so the residual μ_2-class of Round 29's boundary (a) *vanishes identically at the committed model* (Corollary 6.5). Certificate `fold_equation_stack_checks.py`: 36 checks, exit 0, deterministic double-run byte-identical.
- **Q2 (the Clay statement)**: *not claimed*. The one-directional reduction stands (Round 30, Theorem 12 there); the input axis is unchanged. What this round adds is the stratum theorem (Theorem 8.1): within the committed universe, the counterexample locus is now confined to the fold-free stratum — every fold-carried input is discharged end-to-end through the certified stack, the multiplicative layer included.
- **The closing string** (`CONJECTURE IS PROVEN` in the Clay sense): *not emitted*. The only content it could carry at the present state — the assertion that the input-axis items are

discharged — is false; the deck-existence at general input is open, and this document does not override the campaign's crucible.

1 The statement system and the round's position

The two questions are separated as in Round 30: **Q1** is $\mathrm{FS}_\forall$ on the committed world model (every Hodge input (X, p, α) of the model satisfies $m^* > 0$, hence α is algebraic through the committed Theorem N conduit); **Q2** is the Clay statement (every rational (p,p) class on every smooth projective X is rational-algebraic). Round 30 transported the deck–door mechanism to every product world $E \times Y$ at the lattice level and recorded the corpus map; Round 29 (commit `ec1d8b1`) breached the committed-deck boundaries (a)–(d): the uniform kill at every decked input, the honest-invariant dichotomy, the post-strip sector $H^2(G_\bullet; A) = \mathbb{Z}^{12} \oplus (\mathbb{Z}/2)^6$, the gerbe ambient $H^3(G_\bullet; \mu_2) = \mathbb{F}_2^{71}$, and the dressed two-cell — a genuine nonabelian 2-cocycle, 24/24 realizations — with *the residual μ_2-class named as the next strike target.*

Round 31 takes that target, and takes it through the operator's directive: the fold equations themselves, the Hodge structure forced out of them, the counterexamples excluded. The execution is the fold-equation stack of this document: the emergence layer (the fold equation alone forces the Gaussian module structure, the bigrading, the conjugation law, and the door), the forced-kill layer (the section theorem and the split-deck formula: the lifting and the mixed-layer kill are laws of the semidirect structure, not of the committed model), the square-zero dressing law (the multiplicative obstruction quadratic in the cocycle does not exist), and the integral gauge at $2m$ (the doubled transport trivializes *integrally*, in $\mathrm{SL}_8(\mathbb{Z})$, so the μ_2-boundary class vanishes). The stratum theorem assembles the stack; the boundary section states exactly what remains.

Notation. $C_4 = \langle\gamma\rangle$; the periodic resolution gives $H^2(C_4; M) = M^{C_4}/NM$ and $H^1(C_4; M) = \ker N/(\gamma - 1)M$ with $N = 1 + \gamma + \gamma^2 + \gamma^3$. $\mathbb{Z}[i]$ is the Gaussian ring, $N(z) = z\bar{z}$ its norm, $(1 - i)$ the prime over 2 with residue field $\mathbb{Z}[i]/(1 - i) \cong \mathbb{F}_2$. The committed world model is the eight-letter Gaussian lattice world: letters $\mathbb{Z}^8$ with basis $e_0, f_0, e_1, f_1, e_2, f_2, e_3, f_3$, the deck $\gamma = \mathrm{rot}(e_0, f_0) \oplus \mathrm{id}$ (the matrix G of §6.1), the content (Weil) plane $A = \mathbb{Z}W_R \oplus \mathbb{Z}W_I \cong \mathbb{Z}[i]$ with the content rotation CONT, the polarization $\theta = \sum_j e_j \wedge f_j$, and the group $G_\bullet = \Gamma \rtimes C_4$ with $\Gamma \cong \mathbb{Z}[i] \oplus \mathbb{Z}^6$ (one moved pair, six fixed letters). $W_R, W_I = \mathrm{Re}, \mathrm{Im}\,\Omega$, $\Omega = \prod_{j=0}^{3}(e_j + i f_j)$.

2 The fold equation and the emergence layer

Definition 2.1 (The fold equation)**.** *Let M be a finitely generated free $\mathbb{Z}$-module. A* fold *on M is an integral operator $\gamma \in \mathrm{End}(M)$ with $\gamma^2 = -\mathrm{id}_M$. The pair (M, γ) is a* fold lattice*. The* fold equations *are the identities forced on the cohomology of $C_4 = \langle\gamma\rangle$ by this single law.*

2.1 The Gaussian structure theorem

Theorem 2.2 (The Gaussian structure; every fold lattice is a $\mathbb{Z}[i]$-module)**.** *Let (M, γ) be a fold lattice. Then M carries a canonical $\mathbb{Z}[i]$-module structure by $i \cdot m := \gamma m$, and $M \cong \mathbb{Z}[i]^s$ with $s = \frac{1}{2}\operatorname{rank} M$. Conversely every $\mathbb{Z}[i]$-module is a fold lattice for $\gamma =$ multiplication by i.*

Proof. $i\cdot(i\cdot m)=\gamma^2 m=-m=(-1)\cdot m$ and the $\mathbb{Z}$-module axioms are exactly the endomorphism axioms, so the assignment $i\mapsto\gamma$ extends $\mathbb{Z}$-linearly to a ring map $\mathbb{Z}[i]\to\mathrm{End}(M)$. M is torsion-free over $\mathbb{Z}$ by hypothesis; it is torsion-free over $\mathbb{Z}[i]$ because $z\cdot m=0$ with $z=a+bi\neq 0$ gives $N(z)\,m=\bar{z}\,(z\cdot m)=0$, i.e. $(a^2+b^2)m=0$, hence $m=0$. $\mathbb{Z}[i]$ is a Euclidean domain, so the finitely generated torsion-free module M is free: $M\cong\mathbb{Z}[i]^s$, and $\mathrm{rank}_{\mathbb{Z}}\,M=2s$. The converse is immediate. □

Corollary 2.3 (The bigrading; the Hodge skeleton emerges)**.** *For every fold lattice (M,γ):*

$$M\otimes_{\mathbb{Z}}\mathbb{C}\;=\;M^{1,0}\oplus M^{0,1},\qquad M^{1,0}=\ker(\gamma-i),\quad M^{0,1}=\ker(\gamma+i),$$

with $\dim_{\mathbb{C}}M^{1,0}=\dim_{\mathbb{C}}M^{0,1}=s$, complex conjugation swapping the two summands, and γ acting by $+i$ on $M^{1,0}$ and $-i$ on $M^{0,1}$. This is exactly the skeleton of a weight-one Hodge structure: the fold equation forces the bigrading, the conjugation, and nothing else (Remark 2.7).

Proof. $M\cong\mathbb{Z}[i]^s$ by Theorem 2.2; over $\mathbb{C}$, $\mathbb{C}\otimes_{\mathbb{Z}[i]}M\cong\mathbb{C}^s$ splits by the two idempotents $\frac{1}{2}(1\mp i\gamma)$, which is the displayed decomposition. The conjugation $\overline{m\otimes z}$ swaps γ-eigenvalues $\pm i$. The dimensions are equal because $\gamma^2=-1$ makes the eigenvalues paired. □

Theorem 2.4 (The conjugation theorem, universal rational form)**.** *Let (M,γ) be a fold lattice and $0\neq v\in M\otimes\mathbb{Q}$. Then v is not purely one-typed: neither $\gamma v=iv$ nor $\gamma v=-iv$ holds in $M\otimes\mathbb{Q}$.*

Proof. $\gamma v\in M\otimes\mathbb{Q}$; if $\gamma v=\pm iv$ with $v\neq 0$ then $\pm i=\gamma v/v\in\mathbb{Q}$, false. (Equivalently: the minimal polynomial x^2+1 of γ over $\mathbb{Q}$ has no rational root, so γ has no rational eigenvector.) □

This is the universal form of Round 30's conjugation theorem (there: integral classes are never purely $(p-1,p)$-typed; here: the statement is a law of the fold equation on every fold lattice, certificate **A001**).

2.2 The universal door laws

Theorem 2.5 (R1 universal; the norm degeneration and the door group)**.** *Let (M,γ) be a fold lattice. Then:*

1. $M^{C_4}=\ker(\gamma-1)=0$*;*
2. *the norm degenerates:* $N=(1+\gamma)(1+\gamma^2)=0$*;*
3. $H^2(C_4;M)=0$*;*
4. $H^1(C_4;M)=M/(\gamma-1)M$ *is 2-elementary with*

$$H^1(C_4;M)\;\cong\;M/(\gamma-1)M\;\cong\;\big(\mathbb{Z}[i]/(1-i)\big)^s\;\cong\;(\mathbb{Z}/2)^s,$$

and the order formula $|M/(\gamma-1)M|=|\det(\gamma-1)|=N(1-i)^s=2^s$.

Proof. (1) $\gamma v=v\Rightarrow\gamma^2 v=v=-v\Rightarrow 2v=0\Rightarrow v=0$ (torsion-free). (2) $1+\gamma^2=0$ on the nose. (3) $H^2=M^{C_4}/NM=0/0$. (4) The composite $\mathbb{Z}[i]^s\xrightarrow{\times(1-i)}\mathbb{Z}[i]^s$ is exactly $\gamma-1$ up to the unit $-i$ ($i-1=-(1-i)$, $i-1=(-i)(1-i)$), so $M/(\gamma-1)M\cong(\mathbb{Z}[i]/(1-i))^s$; the ideal $(1-i)$ is the unique prime over 2 ($2=-i(1-i)^2$), so $\mathbb{Z}[i]/(1-i)\cong\mathbb{F}_2$ and the group is $(\mathbb{Z}/2)^s$. The index formula: $[\,M:(\gamma-1)M\,]=|\det_{\mathbb{Z}}(\times(1-i))|^s=N(1-i)^s=2^s$. □

The point of Theorem 2.5 for this round: *the door group is the residue field of the Gaussian prime over* 2. The mod-2 structure of the fold universe is not a model accident — it is the ramification of 2 in $\mathbb{Z}[i]$, forced by the fold equation (certificate **A002**, **A003**, **A004**, on randomly conjugated $\mathbb{Z}[i]^s$ decks: the laws hold deck-independently).

Theorem 2.6 (R5 universal; the parity door)**.** *Let (M, γ) be a fold lattice and $m \in M$ an element with nonzero class in $M/(\gamma-1)M \cong (\mathbb{Z}/2)^s$ (e.g. a fold-pair generator, or the committed W_R). Then*

$$k\,m \in (\gamma-1)M \iff k \text{ even.}$$

Proof. Under Theorem 2.5(4) the class of km in $M/(\gamma-1)M \cong (\mathbb{Z}[i]/(1-i))^s$ is k times the class of m; the class of m is nonzero (it is a unit residue along its factor), so the class of km vanishes iff $k = 0$ in $\mathbb{F}_2$, i.e. iff k is even. □

This is the committed door law R5 (the $m^* = 2$ pin of the Weil plane), now proven for every fold lattice — the admission law of the fold equation is the mod-2 class structure of the Gaussian prime (certificate **A005**, **E004**).

Remark 2.7 (The exact scope of the emergence)**.** *What the fold equation forces: the $\mathbb{Z}[i]$-structure, the bigrading, the conjugation law, the door group, the parity door — the complete* lattice skeleton *of a weight-one Hodge structure, universally (family A: six randomly conjugated $\mathbb{Z}[i]^s$ decks, all laws deck-independent). What the fold equation does* not *force: the polarization (the positivity of $\pm\theta$ on the bigraded summands) — positivity is geometry, not lattice algebra, and no claim of this round uses it. The Hodge structures of the committed worlds are carried* through *this skeleton; the skeleton itself is what the forcing argument of §8 uses.*

3 The section theorem: the universal lifting cocycle

Round 28 realized the mixed-cell witness by an explicit lifting cocycle c_ν. This section proves the construction is a law of the semidirect structure itself.

Theorem 3.1 (The section theorem)**.** *Let $G = \Gamma \rtimes C_4$ be a semidirect product with group law $(v,h)(w,k) = (v + \gamma^h w, h+k)$, and let A be a $\mathbb{Z}$-module on which G acts through the deck, $(v,h)\cdot a = \gamma_A^h a$, where $\gamma_A^2 = -\mathrm{id}$ on a 2-torsion-free A. Let $f \in \mathrm{Hom}_{\mathbb{Z}}(\Gamma, A)$ with $Nf = 0$ (the cocycle class: on the deck-fixed letters this is automatic, since there $\gamma_{\mathrm{act}} = \gamma_A$ and $N = 1 + \gamma_A + \gamma_A^2 + \gamma_A^3 = 0$). Define*

$$\nu(h) \;:=\; \sum_{i=0}^{h-1} \gamma_{\mathrm{act}}^i f \;\in\; \mathrm{Hom}(\Gamma, A), \qquad c_\nu\big((v,h),(w,k)\big) \;:=\; \nu(h)\big(\gamma^h w\big),$$

where $\gamma_{\mathrm{act}} f = \gamma_A \circ f \circ \gamma^{-1}$. Then:

1. *ν is a C_4-cocycle ($\delta\nu = 0$) iff $Nf = 0$, the partial-sum definition forcing the in-range telescoping $\nu(h+k) = \nu(h) + \gamma_{\mathrm{act}}^h \nu(k)$ for $h + k \le 4$ and the wraparound $h + k \equiv 0$ being exactly the norm relation $\nu(h) + \gamma_{\mathrm{act}}^h \nu(4-h) = Nf$;*
2. *c_ν satisfies the bar identity: for all $x = (v,h)$, $y = (w,k)$, $z = (u,\ell)$,*

$$\gamma_A^h\, c_\nu(y,z) - c_\nu(xy,z) + c_\nu(x,yz) - c_\nu(x,y) = 0,$$

so c_ν is a genuine 2-cocycle of G with values in A;

3. *the coboundary compatibility:* $c_{\nu+\delta\beta} = c_\nu + \delta(\beta \circ \mathrm{pr}_\Gamma)$ *for every* $\beta \in \mathrm{Hom}(\Gamma, A)$*, where* $\delta\beta(h) = \gamma_{\mathrm{act}}^h \beta - \beta$ *and* $(\delta(\beta \circ \mathrm{pr}))((v,h),(w,k)) = \gamma_A^h \beta(w) - \beta(\gamma^h w)$.

Proof. (1) The in-range case $h + k \le 4$ is the binomial telescoping of the partial sums. For the wraparound $h+k = 4$: $\nu(h)+\gamma_{\mathrm{act}}^h \nu(4-h) = \sum_{i<h} \gamma_{\mathrm{act}}^i f + \gamma_{\mathrm{act}}^h \sum_{j<4-h} \gamma_{\mathrm{act}}^j f = \sum_{i<4} \gamma_{\mathrm{act}}^i f = Nf$, which vanishes by hypothesis; this is also the only relation the C_4-cocycle identity needs beyond the telescoping (the periodic resolution of C_4 has the single norm relation). (2) Expand with $yz = (w + \gamma^k u, k + \ell)$ and $xy = (v + \gamma^h w, h + k)$, using $\gamma^{h+k} = \gamma^{(h+k) \bmod 4}$:

$$c_\nu(x, yz) = \nu(h)\big(\gamma^h w + \gamma^{h+k} u\big) = c_\nu(x, y) + \nu(h)\big(\gamma^{h+k} u\big),$$

so the bar expression collapses to

$$\gamma_A^h \nu(k)\big(\gamma^k u\big) - \nu(h+k)\big(\gamma^{h+k} u\big) + \nu(h)\big(\gamma^{h+k} u\big) = \big[\gamma_{\mathrm{act}}^h \nu(k) + \nu(h) - \nu(h+k)\big]\big(\gamma^{h+k} u\big),$$

where $(\gamma_{\mathrm{act}}^h \nu(k))(\gamma^{h+k} u) = \gamma_A^h\, \nu(k)(\gamma^{-h}\gamma^{h+k} u) = \gamma_A^h \nu(k)(\gamma^k u)$ was used. The bracket vanishes by the cocycle law of (1). (3) Linearity of c in ν reduces this to $c_{\delta\beta} = \delta(\beta \circ \mathrm{pr})$; expanding both sides with the module law $(v,h) \cdot a = \gamma_A^h a$:

$$c_{\delta\beta}((v,h),(w,k)) = \delta\beta(h)(\gamma^h w) = \big(\gamma_{\mathrm{act}}^h \beta - \beta\big)(\gamma^h w) = \gamma_A^h \beta(\gamma^{-h}\gamma^h w) - \beta(\gamma^h w),$$

which is the displayed coboundary. □

Corollary 3.2 (The section property; $d_2 = 0$)**.** *The assignment* $[\nu] \mapsto [c_\nu]$ *of Theorem 3.1(3) descends to cohomology as a* section *of the mixed-cell map* $H^2(G; A) \to H^1(C_4; H^1(\Gamma; A))$*: every mixed class with representative in the image is realized by a lifted two-cell, and the* d_2*-leg of the equivariant-descent spectral sequence vanishes on the post-strip sector. The Round-28 witness construction (*$f_w = \theta(-, e_1)W_R$*,* $Nf_w = 0$ *by the deck-fixed support) is one instance; the construction is class-agnostic.*

The honest scope, on the face: f ranges over the $Nf = 0$ class (the fixed-letter-supported homs; a generic hom on the moved pairs fails the norm relation and ν is not a cocycle at all). Certificate **B001** (five random split decks $(r,t,s) \in \{(1,6,1),(2,4,1),(1,4,2),(3,2,1),(2,2,2)\}$, random fixed-letter-supported f, full bar identity on random triples) and **B002** (the coboundary compatibility).

4 The split-deck formula: the mixed layer as a law of the deck shape

Theorem 4.1 (The split-deck formula)**.** *Let* $\Gamma = \mathbb{Z}[i]^r \oplus \mathbb{Z}^t$ *be a split letter deck (*r *moved pairs,* t *fixed letters;* γ *acts by multiplication by* $-i$ *on the pairs and by the identity on the fixed letters) and* $A \cong \mathbb{Z}[i]^s$ *a 2-torsion-free content module with the content rotation* γ_A*. Then the mixed layer is*

$$H^1\big(C_4; \mathrm{Hom}_{\mathbb{Z}}(\Gamma, A)\big) \;\cong\; (\mathbb{Z}/2)^{ts},$$

pure 2-torsion: each fixed letter contributes exactly the content's door group $(\mathbb{Z}/2)^s$ *(Theorem 2.5), and the* r *moved pairs contribute* nothing *(the pair-block vanishing).*

Proof. $\mathrm{Hom}_{\mathbb{Z}}(\Gamma, A)$ splits as the direct sum of the fixed-letter blocks $\mathrm{Hom}(\mathbb{Z}\ell_j, A) \cong A$ (t copies, the action $\gamma_{\mathrm{act}} = \gamma_A$) and the pair blocks $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[i], A)$ (r copies per Gaussian rank of A). The fixed-letter blocks give $H^1(C_4; A) = (\mathbb{Z}/2)^s$ each by Theorem 2.5(4) — this is the $(\mathbb{Z}/2)^{ts}$.

The pair-block vanishing, in the committed shape ($\mathrm{Hom}(\mathbb{Z}[i], \mathbb{Z}[i]) \cong M_2(\mathbb{Z})$): write the content rotation as $G = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ (multiplication by $-i$ in the basis $(1, i)$) and the inverse letter rotation as $J = G^{-1}$ (multiplication by $+i$). The mixed action on $F \in M_2(\mathbb{Z})$ is the conjugation-type involution

$$T(F) \;=\; G\,F\,G^{-1}, \qquad T^2 = \mathrm{id},$$

so $N_4 = 1 + T + T^2 + T^3 = 2(1 + T)$ and $H^1(C_4; M_2(\mathbb{Z})_T) = \ker(1 + T)/\operatorname{im}(T - 1)$. Computing both sides for $F = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$:

$$T(F) = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}, \qquad (1{+}T)(F) = \begin{pmatrix} a+d & b-c \\ c-b & d+a \end{pmatrix}, \qquad (T{-}1)(F) = \begin{pmatrix} d-a & -(b+c) \\ -(b+c) & -(d-a) \end{pmatrix}.$$

Thus $\ker(1 + T) = P^- = \{\left(\begin{smallmatrix} a & b \\ b & -a \end{smallmatrix}\right)\}$ (the anti-invariants) and $\operatorname{im}(T - 1) = P^-$ *in full* — given a target $\left(\begin{smallmatrix} u & v \\ v & -u \end{smallmatrix}\right)$, take $d = u$, $a = 0$, $c = -v$, $b = 0$. Hence $H^1 = 0$: the pair block contributes nothing. For general (r, s) the same commutator computation runs blockwise on $M_{2s\times 2r}(\mathbb{Z})$ (certificate **C002** computes the vanishing for the grid $(r, s) \in \{(1,1), (2,1), (1,2), (2,2), (3,1)\}$ at $t = 0$; **C001** computes the full formula for the grid $(r, t, s) \in \{(1,6,1), (2,4,1), (1,4,2), (2,2,2), (3,3,1)\}$; the committed mixed layer is the case $(r, t, s) = (1, 6, 1)$, and **E003** re-derives $(\mathbb{Z}/2)^6$ through the universal engine: Smith normal form $[2, 2, 2, 2, 2, 2]$, order 64, free rank 0). □

Theorem 4.2 (The glue control; the $2m$ vs $4m$ boundary)**.** *The split hypothesis is load-bearing: on the non-split glued module $\mathbb{Z}[x]/\big((x+1)(x^2+1)\big)$ (a single letter carrying both an involution part and a rotation part, integrally unsplittable),*

$$H^1\big(C_4;\, \mathbb{Z}[x]/((x+1)(x^2+1))\big) \;\cong\; \mathbb{Z}/4,$$

*so the mixed layer carries genuine 4-torsion and the kill lands at $4m$, not $2m$ (certificate **C001**, **C003**: the companion matrix of $(x+1)(x^2+1)$ computes $H^1 = \mathbb{Z}/4$). This is boundary (d) of Round 28 — the input-dependent glue — exhibited at the universal level:* split decks kill at $2m$; glued decks kill at $4m$; the committed deck is split.

The hostile controls (family D): the trivial deck has $H^2 = (\mathbb{Z}/4)^3 \neq 0$ (the fold operator's $H^2 = 0$ is genuinely the $\gamma^2 = -1$ law, **D001**); the involution deck $\gamma^2 = +1$ has $H^2 = \mathbb{Z}/2 \neq 0$ (**D002**); the negation deck $\gamma = -1$ has $H^2 = 0$ but $H^1 = (\mathbb{Z}/2)^2$, a different signature (**D003**); and the sabotaged formula $(\mathbb{Z}/2)^{ns}$ fails on the committed shape $(n, s) = (8, 1)$ — the correct $(\mathbb{Z}/2)^{ts} = (\mathbb{Z}/2)^6$ is load-bearing (**D004**).

5 The square-zero dressing law

The multiplicative layer of Round 29 dressed the additive cocycle by the envelope $M = I + X(c)$, $X(c)$ the realization matrix of the content $c \in A$. The quadratic obstruction to $c \mapsto I + X(c)$ being a homomorphism of the two-cell calculus is the products $X(c)X(c')$. This section computes the structure of the realization space and proves the obstruction vanishes on every realization span.

Theorem 5.1 (The block law of the rotation lattice)**.** *In the committed model let* $\Phi \in \mathrm{End}(\mathrm{End}(\mathbb{Z}^8))$ *be the deck conjugation* $\Phi(X) = \gamma X \gamma^{-1}$ *and* $L = \ker(\Phi^2 + 1)$ *the rotation lattice. Then, with respect to the* $(2 \mid 6)$ *block split (moved pair | fixed letters) and* $D = \gamma^2 = \mathrm{diag}(-I_2, I_6)$*:*

$$L \;=\; \left\{ \begin{pmatrix} 0 & Q \\ R & 0 \end{pmatrix} \;:\; Q \in M_{2\times 6}(\mathbb{Z}),\; R \in M_{6\times 2}(\mathbb{Z}) \right\}, \qquad \operatorname{rank} L = 24.$$

Proof. $\Phi^2(X) = \gamma^2 X \gamma^{-2} = DXD$. Write $X = \left(\begin{smallmatrix} P & Q \\ R & S \end{smallmatrix}\right)$; then $DXD = \left(\begin{smallmatrix} P & -Q \\ -R & S \end{smallmatrix}\right)$, so $DXD = -X$ iff $P = -P$ and $S = -S$, i.e. $P = 0$ and $S = 0$. The rank is $2 \cdot 6 + 6 \cdot 2 = 24$. (Certificate **F001**: the rank is computed as 24 through the integral kernel of the 64-dimensional system $\Phi^2 + I$.) □

Theorem 5.2 (The equivariant realizations)**.** *Every nonzero* $A' \in L$ *yields the equivariant pair* (A', B') *with* $B' := -\Phi(A')$*, satisfying*

$$\gamma A' \gamma^{-1} = -B', \qquad \gamma B' \gamma^{-1} = A',$$

and $\Phi^2 = -1$ *on* L *makes this a bijection between the elements of* L *and the equivariant envelopes. The* 24 *kernel-basis vectors of Theorem 5.1 (each supported in a single off-diagonal block) give* 24 *realizations (certificate* ***F002****:* 24/24*).*

Proof. $\Phi(B') = \Phi(-\Phi(A')) = -\Phi^2(A') = A'$, and $\Phi(A') = -B'$ by the definition of B'. The basis vectors of $\ker(\Phi^2 + I)$ computed by the integral kernel algorithm are the entry-supported matrices of the block space (the system $P = 0,\ S = 0$ is a coordinate projection), each lying in exactly one off-diagonal block. □

Theorem 5.3 (The square-zero dressing law)**.** *Every equivariant realization span is square-zero: for every realization* (A', B') *of Theorem 5.2,*

$$A'^2 = A'B' = B'A' = B'^2 = 0,$$

and hence $X(c)X(c') = 0$ *for all* c, c' *in the content span of the realization. Consequently the dressing*

$$M(c) \;=\; I + X(c)$$

is a homomorphic lift *of the additive cocycle: the multiplicative two-cell built from* M *carries no quadratic obstruction, and* $\det M(c) = 1$ *(the dressing lives in* SL_8*).*

Proof. Each kernel-basis realization is supported in a single off-diagonal block of Theorem 5.1 (a Q-type matrix $\left(\begin{smallmatrix} 0 & Q \\ 0 & 0 \end{smallmatrix}\right)$ or an R-type matrix), and Φ preserves the block type (the conjugation by $\gamma = \mathrm{diag}(G, I_6)$ acts on Q as $Q \mapsto GQ$ and on R as $R \mapsto RG^{-1}$). So $\mathrm{span}\{A', B'\}$ lies entirely in one off-diagonal block, and the product of any two matrices of the same block type vanishes:

$$\begin{pmatrix} 0 & Q_1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & Q_2 \\ 0 & 0 \end{pmatrix} = 0, \qquad \begin{pmatrix} 0 & 0 \\ R_1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ R_2 & 0 \end{pmatrix} = 0.$$

Thus every product within the span is 0, $X(c)$ is nilpotent with $X(c)^2 = 0$, $(I + X(c))^{-1} = I - X(c)$, and $\det(I + X(c)) = 1$ (unipotent). (Certificate **F003**: 24/24 spans, all four products zero on each; **F005**: $\det = 1$ on span elements.) □

Remark 5.4 (The honest scope of the square-zero law)**.** *The law is a law of the realization* spans*, not of the ambient lattice:* 96 *of the* 576 *basis pairs of* L *have nonzero cross products (the* Q*-*R *pairings* $\left(\begin{smallmatrix} 0 & Q \\ 0 & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & 0 \\ R & 0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} QR & 0 \\ 0 & 0 \end{smallmatrix}\right)$*). Certificate* ***F004*** *receipts this on the face: the ambient* L *is not a square-zero algebra; the dressing algebra — the spans the envelopes actually live on — is. Every use below (Theorem 6.4) is at exactly this scope.*

6 THE INTEGRAL GAUGE AT $2m$ (the strike)

This section is the round's construction. Round 29 built the dressed two-cell — a genuine nonabelian 2-cocycle with values in SL_8 — and named its residual μ_2-class as the next target. Round 31 computes it: *zero*. The mechanism is an integral gauge trivialization at the doubled scale.

6.1 The committed model data

The letters $\mathbb{Z}^8$ with basis $e_0, f_0, e_1, f_1, e_2, f_2, e_3, f_3$ (indices $0, \dots, 7$); the deck

$$\gamma \;=\; G \oplus I_6, \qquad G = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad \text{on } (e_0, f_0),$$

so $\gamma e_0 = -f_0$, $\gamma f_0 = e_0$, and the six letters $e_1, f_1, e_2, f_2, e_3, f_3$ are deck-fixed ($\Gamma \cong \mathbb{Z}[i] \oplus \mathbb{Z}^6$: $r = 1$ moved pair, $t = 6$). The content plane $A = \mathbb{Z}W_R \oplus \mathbb{Z}W_I \cong \mathbb{Z}[i]$ with the content rotation $\mathrm{CONT} = G$ (multiplication by $-i$). The polarization $\theta = \sum_j e_j \wedge f_j$ (the alternating matrix T_θ with $\theta(f_j, e_j) = -1$, $\theta(e_j, f_j) = +1$). The committed witness (Round 28, T4):

$$f_w \;=\; \theta(-, e_1) \otimes W_R : \qquad f_w(v) \;=\; \theta(v, e_1)\, W_R \;=\; -\,v_{f_1}\, W_R,$$

supported on the deck-fixed letter e_1 (so $N f_w = 0$: Theorem 3.1, the fixed-letter norm annihilation). The lifting data ($\nu(h) = \sum_{i<h} \gamma_{\mathrm{act}}^i f_w$, $c_\nu((v,h),(w,k)) = \nu(h)(\gamma^h w)$) are as in §3; $G_\bullet = \Gamma \rtimes C_4$ acts on A through the deck, and on $\mathrm{End}(\mathbb{Z}^8)$ by conjugation $(v,h) \cdot X = \gamma^h X \gamma^{-h}$.

The mixed action on homs is $\gamma_{\mathrm{act}} f = \mathrm{CONT} \circ f \circ \gamma^{-1}$.

6.2 The primitive equation and the integral primitive

Theorem 6.1 (The integral primitive)**.** *Define*

$$\sigma \;\in\; \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}^8, A), \qquad \sigma(v) \;:=\; v_{f_1}\,(W_R - W_I)$$

(the f_1-coefficient times the anti-invariant content vector). Then

$$(\gamma_{\mathrm{act}} - 1)\,\sigma \;=\; 2\, f_w,$$

an exact identity of homs. In particular $2f_w$ is an integral $(\gamma_{\mathrm{act}} - 1)$-coboundary, while f_w itself is not (Theorem 6.2): the halves obstruction dies exactly at scale 2.

Proof. Both σ and f_w read the single coefficient v_{f_1} (the index-3 letter, deck-fixed: the moved pair (e_0, f_0) is disjoint from f_1, so $(\gamma^{-1}v)_{f_1} = v_{f_1}$). With $\mathrm{CONT} = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ in the basis (W_R, W_I):

$$(\gamma_{\mathrm{act}}\sigma)(v) = \mathrm{CONT}\,\sigma(\gamma^{-1}v) = v_{f_1}\,\mathrm{CONT}(W_R - W_I) = v_{f_1}\,\mathrm{CONT}(1,-1) = v_{f_1}\,\big(-W_R - W_I\big),$$

so

$$(\gamma_{\mathrm{act}}\sigma - \sigma)(v) = v_{f_1}\big[-(W_R + W_I) - (W_R - W_I)\big] = -2\,v_{f_1} W_R \;=\; 2\, f_w(v),$$

using $f_w(v) = -v_{f_1} W_R$ ($\theta(f_1, e_1) = -1$). (Certificate **G001**: the identity is verified on all eight letters; the committed signs are as displayed.) □

Theorem 6.2 (The scale-1 halves obstruction; the hostile)**.** *f_w is* not *an integral $(\gamma_{\mathrm{act}} - 1)$-coboundary: the scale-1 transport is not trivializable, and the kill at $2m$ is load-bearing (certificate **G006**: the 16-dimensional Hom system has no integral solution for $(\gamma_{\mathrm{act}} - 1)\tau = f_w$; the obstruction is the halves class of the door group, cf. Theorem 2.5).*

6.3 The additive primitive

Theorem 6.3 (The additive primitive; $\delta s = 2c_\nu$)**.** *Let $s\colon G_\bullet \to A$ be the 1-cochain $s((v,h)) = \sigma(v)$ and $\delta s((v,h),(w,k)) = \gamma_A^h s(w,k) - s\big((v,h)(w,k)\big) + s(v,h)$ the group coboundary. Then*

$$\delta s \;=\; 2\,c_\nu,$$

i.e. the doubled additive two-cell is an exact integral coboundary through the primitive σ.

Proof. Since σ reads a deck-fixed coefficient, s is constant on the deck coordinate; with $xy = (v+\gamma^h w, h+k)$ and σ linear:

$$\delta s(x,y) = \mathrm{CONT}^h\sigma(w) - \sigma(v+\gamma^h w) + \sigma(v) = \mathrm{CONT}^h\sigma(w) - \sigma(\gamma^h w).$$

Define $T_h(w) := \sigma(\gamma^h w) - \mathrm{CONT}^h\sigma(w)$; the claim is $T_h(w) = -2c_\nu(h,w)$ for all $h \in \{0,1,2,3\}$, $w \in \mathbb{Z}^8$.

- $h = 0$: $T_0 = 0 = -2c_\nu(0,\cdot)$ (as $\nu(0) = 0$).
- $h = 1$: by Theorem 6.1 applied at the argument γw, $\mathrm{CONT}\,\sigma(w) - \sigma(\gamma w) = 2f_w(\gamma w)$, so $T_1(w) = -2f_w(\gamma w) = -2c_\nu(1,w)$ (as $c_\nu(1,w) = \nu(1)(\gamma w) = f_w(\gamma w)$).
- Induction $h \mapsto h+1$ (indices mod 4):

  $$T_{h+1}(w) = \sigma(\gamma^{h+1}w) - \mathrm{CONT}^{h+1}\sigma(w) = T_h(\gamma w) + \mathrm{CONT}^h\big[\sigma(\gamma w) - \mathrm{CONT}\sigma(w)\big] = T_h(\gamma w) + \mathrm{CONT}^h T_1(w).$$

  Using the expansion $c_\nu(h,w) = \sum_{i<h} \mathrm{CONT}^i f_w(\gamma^{h-i}w)$ (which is $\nu(h)(\gamma^h w)$ with $(\gamma^i_{\mathrm{act}} f_w)(u) = \mathrm{CONT}^i f_w(\gamma^{-i}u)$), the inductive step gives

  $$T_{h+1}(w) = -2\sum_{i<h} \mathrm{CONT}^i f_w(\gamma^{h+1-i}w) \;-2\,\mathrm{CONT}^h f_w(\gamma w) = -2\sum_{i\le h} \mathrm{CONT}^i f_w(\gamma^{h+1-i}w) = -2\,c_\nu(h{+}1,w).$$

□

(Certificate **G002**: random triples and all structured letter cells, $h,k \in 4\times 4$.)

6.4 The gauge theorem

Now the multiplicative layer. Fix an equivariant realization (A', B') (Theorem 5.2) and write the content dressing

$$X(c) \;:=\; c_R\,A' + c_I\,B' \;\in\; \mathrm{End}(\mathbb{Z}^8) \qquad \big(c = c_R\,W_R + c_I\,W_I \in A\big),$$

so that $\gamma X(c)\gamma^{-1} = X(\mathrm{CONT}\,c)$ (the intertwining law: $\gamma A'\gamma^{-1} = -B'$, $\gamma B'\gamma^{-1} = A'$), $X(c)X(c') = 0$ on the span (Theorem 5.3), and $M(c) = I + X(c) \in \mathrm{SL}_8(\mathbb{Z})$.

Theorem 6.4 (The integral gauge at $2m$)**.** *Define the gauge*

$$S\colon G_\bullet \to \mathrm{SL}_8(\mathbb{Z}), \qquad S\big((v,h)\big) \;:=\; I + X\big(\sigma(v)\big),$$

and the doubled dressed transport two-cell

$$M_2\big((v,h),(w,k)\big) \;:=\; I + X\big(2\,c_\nu\big((v,h),(w,k)\big)\big).$$

Then

$$\delta S \;=\; M_2,$$

where the nonabelian coboundary is

$$(\delta S)(x,y) \;=\; S(x)\cdot\big(\gamma^h S(y)\gamma^{-h}\big)\cdot S(xy)^{-1}$$

($x=(v,h)$, $y=(w,k)$, the $G_\bullet$-action on SL_8 through the deck). The gauge is normalized ($S(1)=I$), integral, and $\det S=1$: the doubled dressed transport is exactly *gauge-trivial in the integral structure group.*

Proof. By the intertwining law, $\gamma^h S(y)\gamma^{-h} = I + X(\mathrm{CONT}^h\sigma(w))$; by the square-zero law, $(I+X(c))^{-1} = I - X(c)$ and all cross products of the X-terms vanish. Hence

$$\begin{aligned}(\delta S)(x,y) &= \big(I+X(\sigma(v))\big)\,\big(I+X(\mathrm{CONT}^h\sigma(w))\big)\,\big(I-X(\sigma(v)+\gamma^h w)\big)\\ &= I + X\big(\sigma(v)+\mathrm{CONT}^h\sigma(w)-\sigma(v)-\sigma(\gamma^h w)\big) \qquad \text{(square-zero kills the quadratic terms)}\\ &= I + X\big(\mathrm{CONT}^h\sigma(w)-\sigma(\gamma^h w)\big) = I + X\big(2\,c_\nu(x,y)\big) = M_2(x,y),\end{aligned}$$

the penultimate step being Theorem 6.3 (whose proof shows $\mathrm{CONT}^h\sigma(w)-\sigma(\gamma^h w) = -T_h(w) = 2c_\nu$). The difference of the two sides of $\delta S = M_2$ is affine-linear in (v,w) for fixed (h,k) (all quadratic terms vanish identically by the square-zero law), so the *complete basis sweep* — $v\in\{0,e_0,\dots,e_7\}$, $w\in\{e_0,\dots,e_7\}$, $(h,k)\in 4\times 4$: 1152 cells — closes the universal quantifier (certificate **G003**). The identity is uniform in the realization: **G004** runs the random battery on all 24 realizations. Integrality and normalization are σ integral, A', B' integral, $\sigma(0)=0$ (**G005**). □

Corollary 6.5 (The residual μ_2-class is zero; the Round-29 target discharged)**.** *The doubled dressed transport M_2 is an integral coboundary in $\mathrm{SL}_8(\mathbb{Z})$, so its nonabelian class $[M_2]\in H^2(G_\bullet;\mathrm{SL}_8)$ vanishes, and the residual μ_2-class of the doubled dressed transport — the boundary-(a) object that Round 29 named as the next strike target, the class measured in the gerbe ambient $H^3(G_\bullet;\mu_2)=\mathbb{F}_2^{71}$ — is* computed: zero. *The reconstruction chain of the committed model closes through the multiplicative layer: boundary (a) of Round 28 is discharged at the committed deck.*

The falsifiability receipts: the trivial gauge $S=I$ fails on the witness cell ($\delta I = I \neq M_2$; certificate **G007**), and the scale-1 gauge fails by Theorem 6.2 (**G006**) — the doubling is not decoration; it is exactly what kills the halves class.

7 The μ_2 layer and the gerbe ambient

Theorem 7.1 (The mod-2 universe of the deck)**.** *$H^p(C_4;\mu_2)=\mu_2$ for every $p\geq 1$: the entire central cohomology of the deck is 2-torsion. Computed genuinely (certificate **H001**, $\mathbb{F}_2$ Gaussian elimination on the action matrices): with $D=\gamma-1$ and $N_4 = 1+\gamma+\gamma^2+\gamma^3$ over $\mathbb{F}_2$, $H^{2k}=\ker D/\operatorname{im} N_4$ and $H^{2k+1}=\ker N_4/\operatorname{im} D$, and the battery computes $\mu_2\to 1$ in every degree, the swap block $P\to 1$, the letter module ($G \bmod 2$, the (e_0,f_0) transposition) $\to 7$, the order-4 unipotents $\to 1$ (size 3) and $\to 0$ (size 4, $N_4=N^3\neq 0$ — the hostile: the computation discriminates).*

Theorem 7.2 (The gerbe ambient recomputed)**.** *The split total complex of the mod-2 letter deck computes*

$$\dim_{\mathbb{F}_2} H^n(G_\bullet;\mu_2) = \sum_{q=0}^{n} d(q), \qquad d(q) = \binom{6}{q} + \binom{6}{q-1} + \binom{6}{q-2},$$

so $\dim H^1 = 8$, $\dim H^2 = 30$, $\dim H^3 = 71$*: the gerbe ambient* $H^3(G_\bullet;\mu_2) = \mathbb{F}_2^{71}$ *of Round 29 is reproduced. The computation is genuine (certificate* ***H002****): the wedge matrices of the mod-2 letter action (the* (e_0, f_0) *transposition* τ*; over* $\mathbb{F}_2$ *the wedge sign is* $+1$*) are built in the subset basis (dims* $1/8/28/56$*), their* C_4*-cohomology is computed by elimination (*$d(q) = 1, 7, 22, 41$ *for* $q = 0, 1, 2, 3$*, matching the binomial decomposition into trivial and swap copies), and the assembly sums the split degrees. With Corollary 6.5 the residual class* itself *is computed: zero.*

The structure of the ambient: every μ_2-obstruction of the C_4 universe is 2-torsion (Theorem 7.1), so the doubling engine kills the obstruction layer as such; and Theorem 6.4 computes the stronger statement: the doubled transport itself is integrally trivial, its gauge living in $\mathrm{SL}_8(\mathbb{Z})$, with the scale-1 halves obstruction receipting (Theorem 6.2) that the doubling is exactly what carries the kill.

8 The stratum theorem: the forcing argument

The operator's directive: force the Hodge structure out of the fold equations; that alone excludes the counterexamples. The executed form, at exactly the strength the certificates support:

Theorem 8.1 (The stratum theorem)**.** *In the committed world model, the fold-carried stratum — the sector of inputs whose reconstruction chain is carried by the fold equation's deck machinery — contains no counterexample:*

1. ***The additive chain*** *(Rounds 25–28, universalized this round): the fold equation forces the door group* $(\mathbb{Z}/2)^s$ *and the parity door (Theorems 2.5, 2.6); the lifting is a law of the semidirect structure (Theorem 3.1); the mixed layer is* $(\mathbb{Z}/2)^{ts}$ *with the pair-block vanishing (Theorem 4.1); the witness class is nonzero at scale* 1 *and dies at scale* 2 *(certificate* ***I001****).*

2. ***The multiplicative chain*** *(this round): the dressing is a homomorphic lift — no quadratic obstruction (Theorem 5.3); the doubled dressed transport is integrally gauge-trivial in* $\mathrm{SL}_8(\mathbb{Z})$ *(Theorem 6.4); the residual* μ_2*-class vanishes identically (Corollary 6.5).*

3. ***The conduit closes at the kill scale****: the kill scale is computed* $m_{\mathrm{kill}} = 2$ *(the door solve, certificate* ***I002****); the two-center conduit constant at* $p = 2$ *is* $(p-1)!\, m_{\mathrm{kill}} = 2$*, and the halving* $\frac{1}{2}$ *lands in* $\mathrm{CH}^2_{\mathbb{Q}}$ *exactly — the rational statement closes at the kill scale, and the uniform* 16 *bound covers every decked input (the Round-29 uniform kill; the scale chain* $\{1 :$ *refused,* $2 :$ *the committed kill,* $16 :$ *the uniform bound*$\}$ *computed through the same integral solver).*

Therefore every fold-carried input is discharged end-to-end — additively, multiplicatively, and through the conduit — and the counterexample locus of the committed universe is confined to the fold-free stratum*: the input axis (ledger 085), where the* C_4*-deck structure itself is the open object.*

Proof. The certified chain composes: (1) is the certified stack of Rounds 25–28 with the universalization of §2–4 (the lattice-level laws are deck-independent, so the chain's group-theoretic layers hold on every fold lattice of the committed shape); (2) is Theorems 5.3 and 6.4 with their certificates; (3) is the computed scale chain of **I002** with the committed two-center conduit (Lemma N of the law stack: $c_p(E^+ - E^-) = (-1)^{p-1}(p-1)!\, m\, \alpha$; at $p = 2$, $m = m_{\text{kill}} = 2$ the rational reconstruction needs exactly the coefficient $\frac{1}{2}$). An input carried by the fold machinery traverses all three chains; each chain is certified at its scale; the composition leaves no undischarged obstruction in the stratum. The confinement statement is the contrapositive: a counterexample in the committed universe must live where the fold machinery does not carry the input — the fold-free stratum. □

Remark 8.2 (What the stratum theorem is and is not)**.** *The theorem is a statement about the committed world model (Q1's universe): the fold-carried sector is certified counterexample-free, the multiplicative layer included — this is the round's content, and it is the honest form of the forcing directive. It is* not *the Clay statement: Q2 quantifies over every variety with no committed structure, and the fold-carried stratum of the committed universe is a proper part of that world. The stratum theorem sharpens the boundary: what separates this program from Q2 is exactly the fold-free stratum — the input axis — and nothing else in the certified chain.*

9 The boundary (on every face)

Theorem 9.1 (The boundary after Round 31)**.**

1. ***The input axis (ledger 085; the single open object)****: the deck-existence at general input. The fold equation's laws are universal (this round), the committed chain closes through the multiplicative layer (this round), but the C_4-deck structure on the input's own cohomology — outside the committed and product worlds — is not constructed. Items (i)–(iii) of the reduction theorem (Round 30, Theorem 12) live here.*

2. ***The discharged boundaries at the committed model****: (a) the residual μ_2-class — computed zero (Corollary 6.5); the gerbe ambient $\mathbb{F}_2^{71}$ stands, the class in it vanishes. (b) The fiber-centralizer condition and the honest-invariant dichotomy — Round 29's discharge, unchanged. (d) The glue boundary — now exhibited at the universal level (Theorem 4.2: split decks kill at $2m$, glued at $4m$); the committed deck is split.*

3. ***The falsifiable line (unchanged)****: exhibit the C_4-deck structure at a general input; reduce the two-cell class there; exclude the non-scalar unexchanged content with a (p,p)-sensitive step. Any one of the three is the next theorem of this program; the stratum theorem makes precise what such a theorem must add: a fold on the input axis.*

The closing statement

Everything this round's mathematics supports, stated at exactly its scope:

- **The fold-equation stack is certified**: the emergence layer (every fold lattice is a $\mathbb{Z}[i]$-module; the bigrading, the conjugation theorem, the door group, the parity door — laws of the fold equation alone); the section theorem (the lifting cocycle is a law of any semidirect C_4-extension); the split-deck formula $(\mathbb{Z}/2)^{ts}$ with the pair-block vanishing and the $\mathbb{Z}/4$

glue control; the square-zero dressing law (the multiplicative two-cell carries no quadratic obstruction, at the receipted per-span scope); **the integral gauge at** $2m$ ($\delta S = M_2$ with $S\colon G_\bullet \to \mathrm{SL}_8(\mathbb{Z})$ integral, normalized, the 1152-cell sweep closing the universal quantifier, uniform over all 24 realizations, the scale-1 halves obstruction receipting that the kill is load-bearing); the μ_2 layer ($H^p(C_4;\mu_2) = \mu_2$; the ambient $\mathbb{F}_2^{71}$ recomputed genuinely).

- **The residual μ_2-class of the doubled dressed transport is computed: zero.** Round 29's named next-strike target is discharged at the committed model; the reconstruction chain closes through the multiplicative layer.
- **The stratum theorem**: no counterexample exists in the fold-carried stratum of the committed universe; the counterexample locus is confined to the fold-free stratum — the input axis. Q1 stands, extended through the multiplicative layer; **Q2 is not claimed**; the one-directional reduction stands; the closing string is not emitted, because the only content it could carry at the present state is the false assertion that the input-axis items are discharged. The next theorem of this program is a fold on the input axis.

ENGINE (per SKILL §35)

The construction engines this round: the fold equation itself ($\gamma^2 = -1$ on the 2-torsion-free lattice — the single law from which the Gaussian module structure, the bigrading, the door group, and the parity door are forced); the telescoping lifting engine $\nu(h) = \sum_{i<h} \gamma_{\mathrm{act}}^i f$ (the section theorem); the block-conjugation engine of the pair-block vanishing; the off-diagonal block engine of the rotation lattice (the square-zero law); and the primitive-gauge engine $\sigma(v) = v_{f_1}(W_R - W_I)$, $S = I + X(\sigma)$ (the integral gauge at $2m$ — the round's strike).

Statement integrity (per SKILL §25)

Diffed against the operator's order (the fold equations themselves; the Hodge structures emerge from the fold equations, forcing it; that alone gives that no counterexample exists): this document delivers the emergence layer in full (the fold equation forces the Hodge skeleton — the $\mathbb{Z}[i]$-structure, the bigrading, the conjugation law, the door — universally), the forcing executed through the certified stack, and the counterexample exclusion in its honest form (the stratum theorem: the fold-carried stratum is counterexample-free; the locus is confined to the fold-free stratum). The exclusion is stated at the committed universe's scope, not the Clay scope, because that is the scope the certificates support; the polarization is not claimed to emerge (Remark 2.7); the square-zero law is stated at its per-span scope; the general-input deck remains open and is on the face. No silent strengthening, no silent weakening, no substitution of scope.

A The certificate inventory (36 checks, exit 0)

`fold_equation_stack_checks.py`: exact arithmetic (`Fraction`/`int`), zero floats, zero randomness beyond fixed LCG seeds, deterministic double-run byte-identical, hostile controls throughout. The families:

Check	Content
A000	Six random $\mathbb{Z}[i]^s$ fold decks constructed ($\gamma^2 = -I$ verified; unimodular conjugates of the standard complex structure).
A001	The fold bigrading on every random deck: no rational eigenline (the universal conjugation theorem, rational form).
A002	R1 universal: $N = 0$, $\ker(\gamma - 1) = 0$, $H^2(C_4; M) = 0$ on every random deck.
A003	$H^1(C_4; M) = (\mathbb{Z}/2)^s$ on every random deck (the door group).
A004	The order formula $\lvert\det(\gamma - 1)\rvert = 2^s$ on every random deck.
A005	R5 universal: $k\,e_0 \in \mathrm{im}(\gamma - 1)$ iff k even, on the random deck.
B001	The section theorem: the bar identity of c_ν on random split decks (fixed-letter-supported f, the $Nf = 0$ class).
B002	The coboundary compatibility $c_{\nu+\delta\beta} = c_\nu + \delta(\beta \circ \mathrm{pr})$.
C001	The split-deck formula $H^1 = (\mathbb{Z}/2)^{ts}$ on the grid $(r, t, s) \in \{(1,6,1), (2,4,1), (1,4,2), (2,2,2), (3,3,1)\}$ (the committed shape re-derived through the universal engine).
C002	The pair-block vanishing on the grid $(r, s) \in \{(1,1), (2,1), (1,2), (2,2), (3,1)\}$ ($t = 0$: $H^1 = 0$).
C003	The glue control: $H^1 = \mathbb{Z}/4$ on $\mathbb{Z}[x]/((x+1)(x^2+1))$ (the $2m$ vs $4m$ distinction).
D001	Hostile trivial deck: $H^2 = (\mathbb{Z}/4)^3 \neq 0$, $H^1 = 0$.
D002	Hostile involution deck ($\gamma^2 = +1$): $H^2 = \mathbb{Z}/2 \neq 0$.
D003	Hostile negation deck ($\gamma = -1$): $H^2 = 0$, $H^1 = (\mathbb{Z}/2)^2$.
D004	Sabotage: $(\mathbb{Z}/2)^{ns}$ fails on the committed shape (the correct $(\mathbb{Z}/2)^{ts}$ is load-bearing).
E001	The committed deck anchor: $\gamma^4 = I$; $\gamma^2 = -1$ on the moved pair, $+1$ on the six fixed letters.
E002	The content rotation $\mathrm{CONT}^2 = -I$ (the Weil plane).
E003	The committed mixed layer $(\mathbb{Z}/2)^6$ re-derived (SNF $[2]^6$, order 64).
E004	R5 on the committed Weil plane: mW_R admitted iff m even (the $m^* = 2$ pin).
E005	$\gamma^3 = \gamma^{-1}$ (the integral inverse; C_4).
F001	The rotation lattice $L = \ker(\Phi^2 + 1)$ has rank 24.
F002	24 equivariant realizations ($\gamma A' \gamma^{-1} = -B'$, $\gamma B' \gamma^{-1} = A'$).
F003	The square-zero dressing law: 24/24 spans, all products zero.
F004	The honest scope receipt: 96 of 576 lattice-wide cross pairs nonzero (the law is per-span).
F005	$\det(I + X(c)) = 1$ on span elements (the dressing in SL_8).
G001	The integral primitive: $(\gamma_{\mathrm{act}} - 1)\sigma = 2f_w$ on all eight letters.
G002	The additive primitive: $\delta s = 2c_\nu$ (random triples + structured cells).
G003	The complete basis sweep: $\delta S = M_2$ on all 1152 cells (the universal quantifier closed).
G004	The gauge identity on all 24 realizations (random battery each).
G005	The gauge normalized and integral: $S\colon G_\bullet \to \mathrm{SL}_8(\mathbb{Z})$.
G006	Hostile scale-1: f_w not an integral coboundary (the halves obstruction; the kill at $2m$ load-bearing).
G007	Sabotage: the trivial gauge fails on the witness cell.
H001	$H^p(C_4; \mu_2) = \mu_2$ every p (genuine $\mathbb{F}_2$ battery: $\mu_2 \to 1$, $P \to 1$, letters $\to 7$, unipotents $\to 1/0$).

Check	Content
H002	The gerbe ambient recomputed (wedge matrices, elimination, $d(q) = 1, 7, 22, 41$; dims 8/30/71).
I001	The stratum receipt: the mixed class nonzero at 1, dead at 2.
I002	The conduit at the kill scale: $m_{\text{kill}} = 2$ computed; the halving lands at denominator 2; the scale chain $\{1, 2, 16\}$.